Level 9 · about 40 h · Python 3.8+

Project: A small Bag whose len is the number of items you stored
Examples run on CPython 3.12.3. **You will be able to:**

- Call len and see it use the __len__ method
- Name two PEPs this lab already cites for dates and style
- Tell a style guide apart from a language rule
- Find the data model chapter in the language reference

PEPs and the Data Model

len asks the object

The data model says operations such as `len` look up special methods. A class can supply `__len__`. `len` then returns that integer.

```
class Bag:
    def __init__(self, items):
        self.items = list(items)

    def __len__(self):
        return len(self.items)

print(len(Bag(["pen", "ink"])))
print("py".__len__())
```

```
Output (CPython 3.12.3):
2
2
```

A PEP is a design document

PEP 8 is the style guide. Formatting tools may warn about it. CPython does not reject a program for ignoring it. PEP 790 is the 3.15 release schedule. This lab's sources record 3.15.0 final as 9 October 2026 from that PEP. The PEP index lists the documents.

```
cited = ["PEP 8", "PEP 790"]
print(", ".join(cited))
print("style guide" != "syntax rule")
```

```
Output (CPython 3.12.3):
PEP 8, PEP 790
True
```

Where to read next

Open the data model chapter when you want the rule for a special method. Open the PEP index when you want the history of a change. The pages linked under Sources are the ones this Level uses.

Check yourself

1. Which method does `len` call?
2. Is PEP 8 a compiler rule?
3. Which PEP in the sources is the 3.15 schedule?

Sources

- Python reference: Data model
- PEP 8 – Style Guide for Python Code
- PEP 790 – Python 3.15 Release Schedule
- PEP index

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